



Sustainable Engineering 3: Design for Environment

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Design for Environment



- Outline the key components of the process of design for environment
- Present examples of using this method for the assessment of design of materials
- Apply the process of design for environment to a selected product

Design for Environment

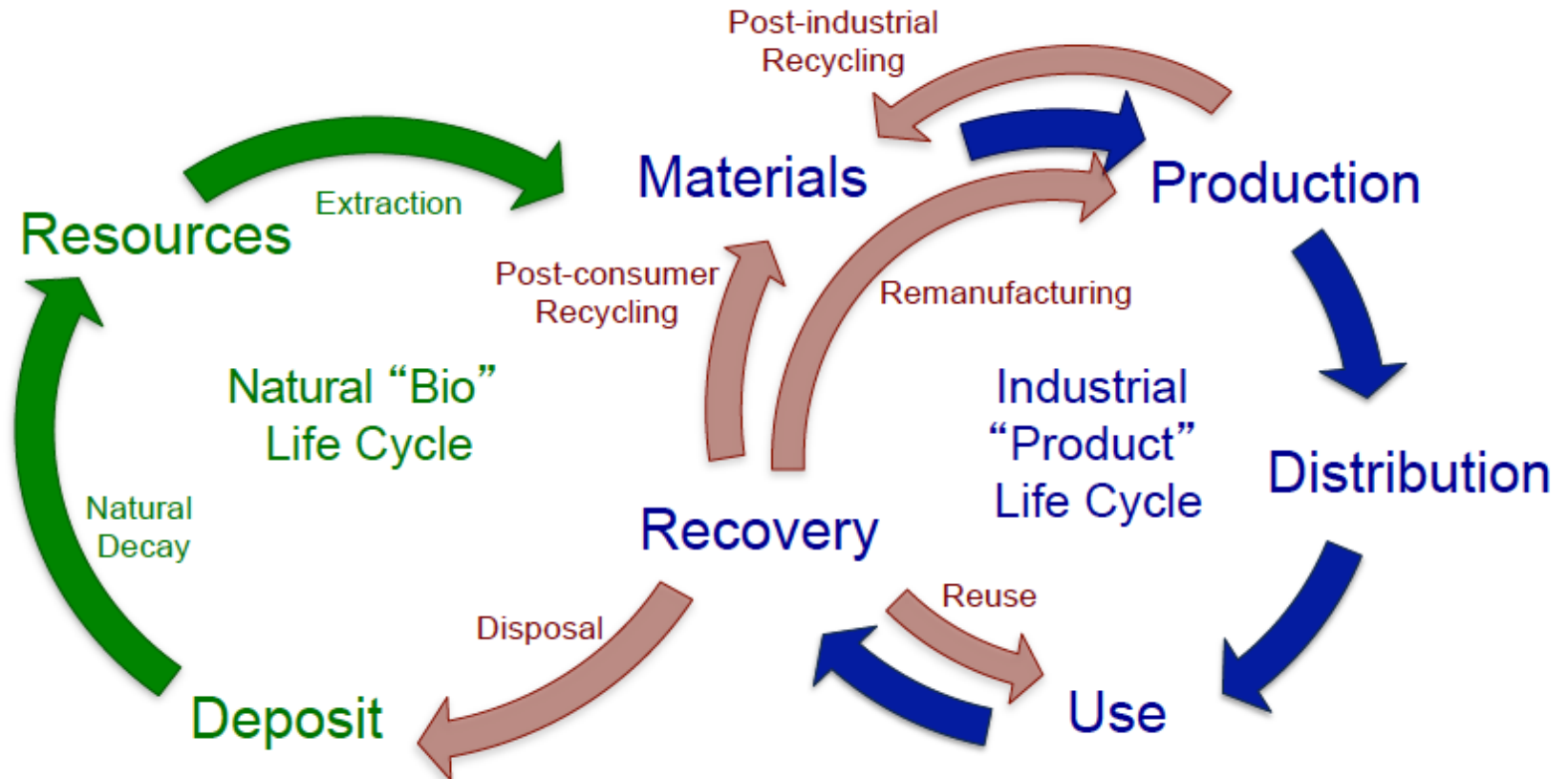


‘Design for Environment (DfE) is a field of product design methodology that includes tools, methods and principles to help designers reduce environmental impact’

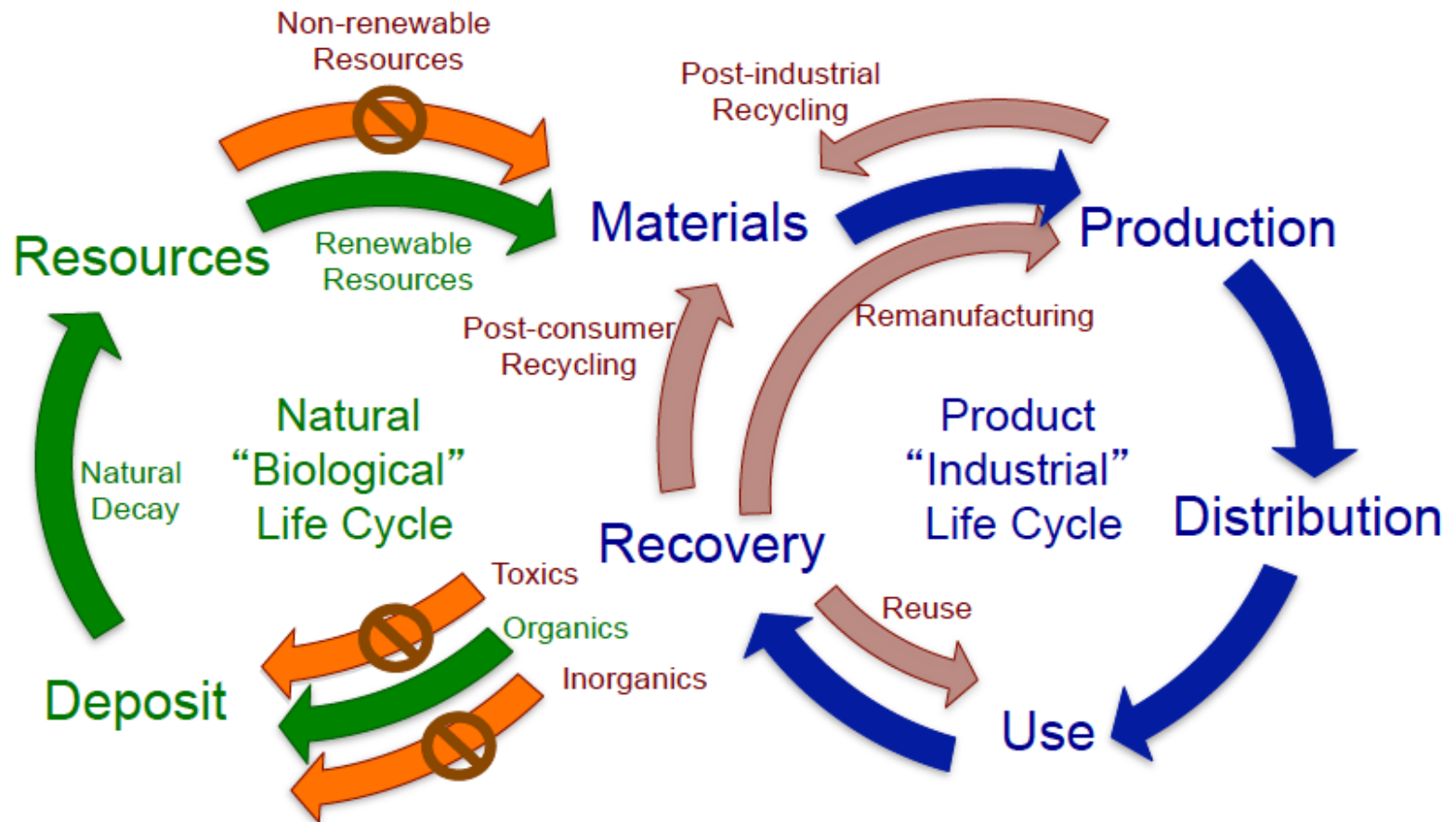
(Telenko et al, 2008)



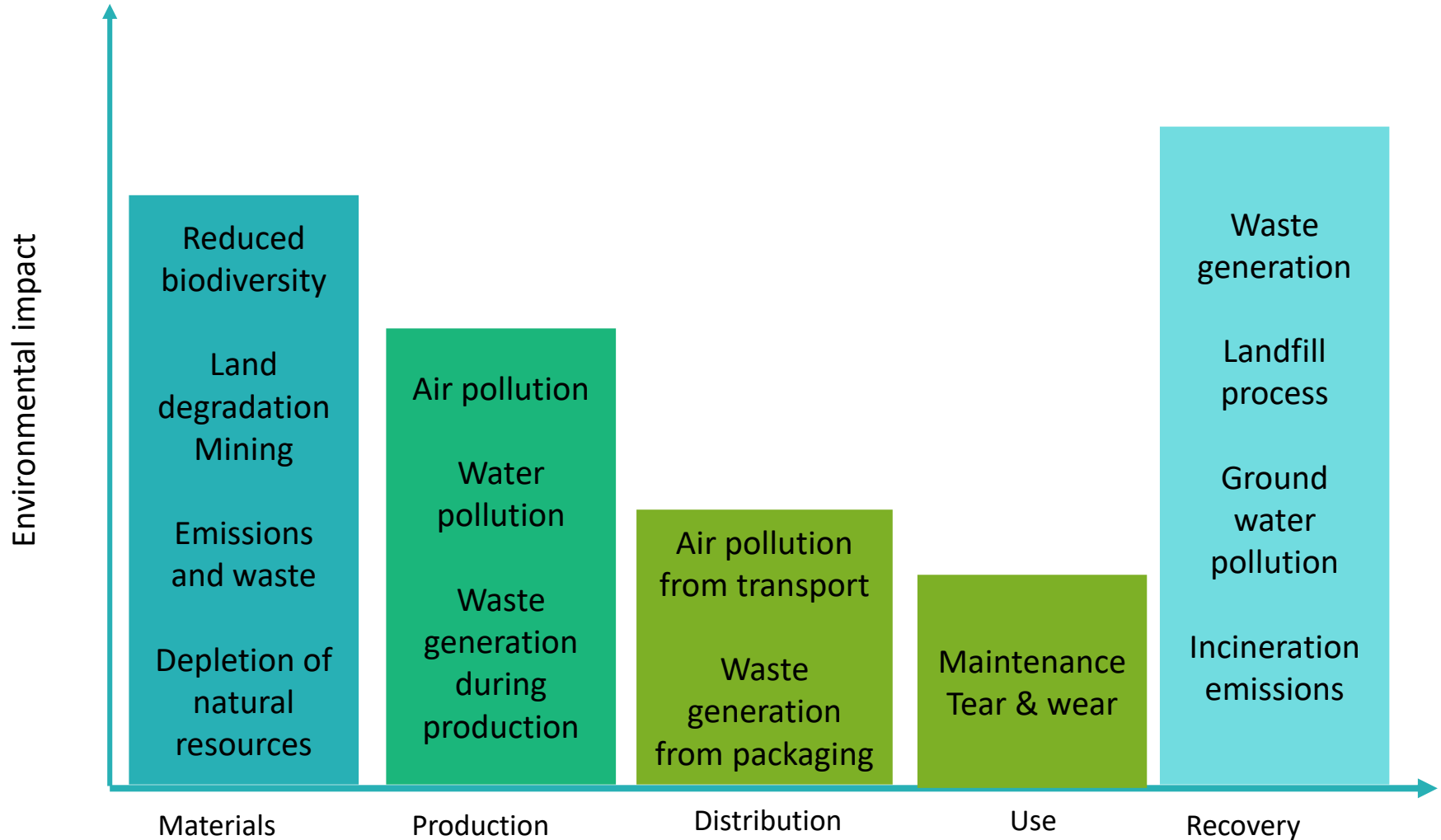
Two life cycles



Limitations to cycle continuity



Environmental impact



Environmental impact



How to assess environmental impact?
A qualitative approach

- What significant sources of impact are in each life cycle stage?
- How 'bad' is the impact?
- Use of less detailed information
- Ranking according to importance given by assessor

Environmental impact

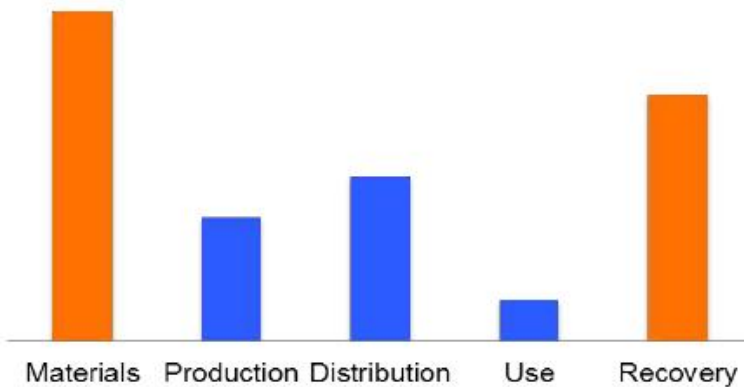
Questions for
consideration
during
qualitative
evaluation

Life Cycle Stage	Questions
Materials	• How much, and what types of plastic will be used? • How much, and what types of additives will be used? • How much, and what types of metals will be used? • How much, and what other types of materials will be used (glass, ceramics etc.)? • What is the environmental profile of these materials? • How much energy is will be required to extract these materials? • Which means of transport will be used to procure them?
Production	• How many, and what types of production processes will be used? • How much, and what types of auxiliary materials are needed? • How high will the energy consumption be? • How much waste will be generated? • Can production waste be separated for recycling?
Distribution	• What kind of transport packaging, bulk packaging and retail packaging will be used (volumes, weights, materials, reusability)? • Which means of transport will be used?
Use	• How much, and what type of energy will be required? • How much, and what kind of consumables will be needed? • What will be the technical lifetime? • How much maintenance and repairs will be needed? • What and how much auxiliary materials and energy will be required? • What will be the aesthetic lifetime of the product?
Recovery	• How will the product be disposed? • Can the product be quickly disassembled using common tools? • Will the components or materials be reused? • Can reusable components be disassembled without damage? What materials will be recyclable? • Will the materials be identifiable?

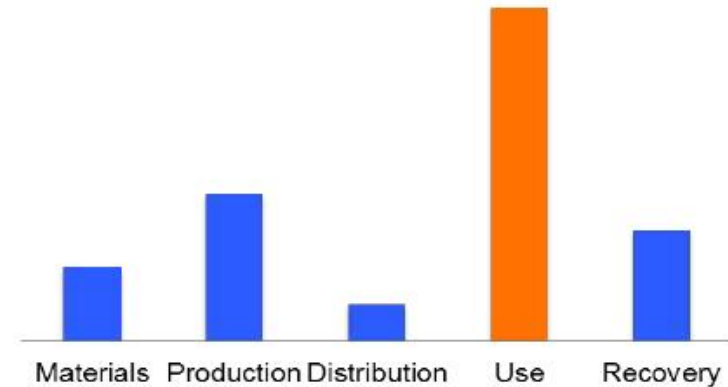
Life cycle impact



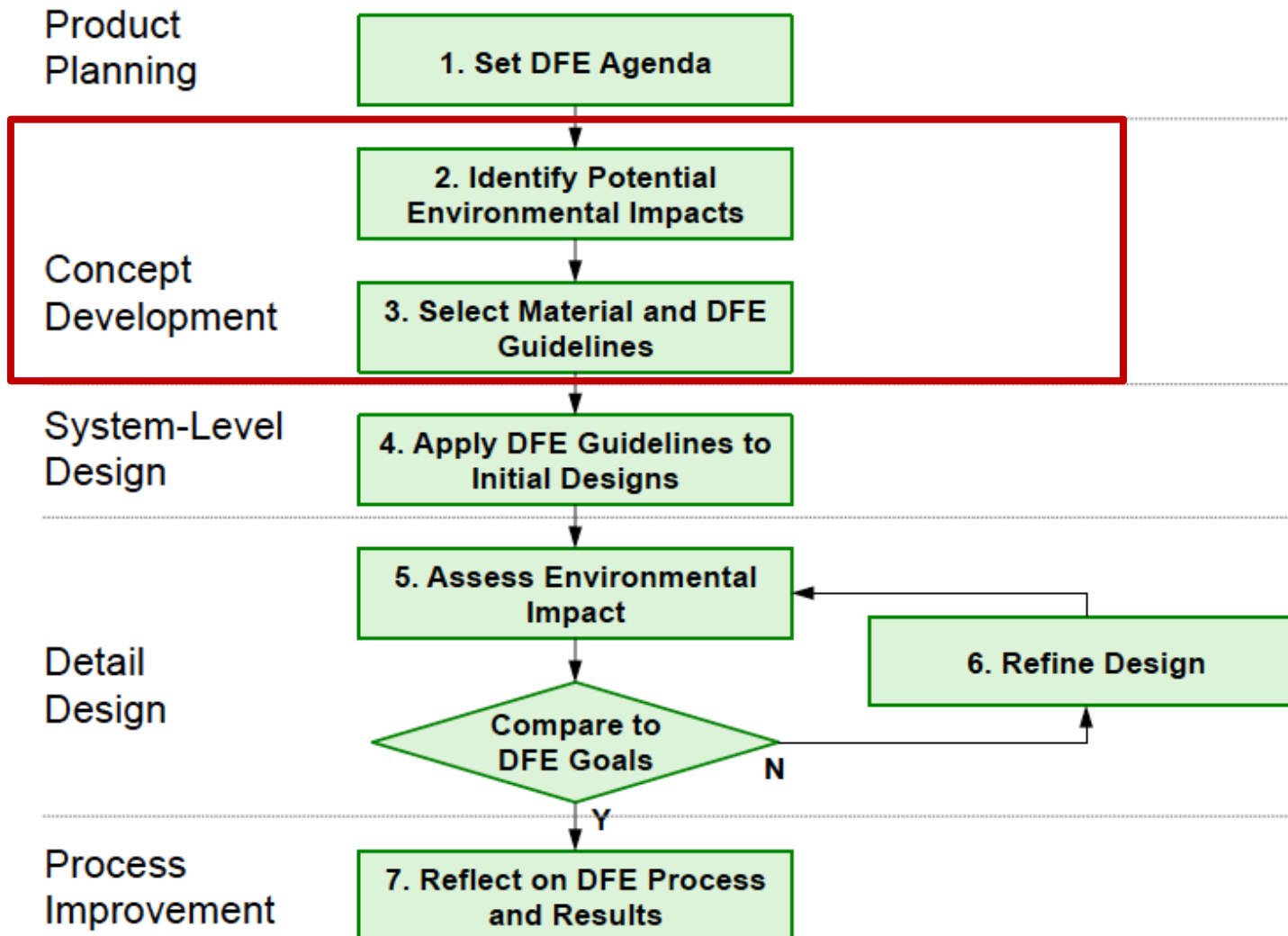
Materials-Intensive Product Impact



Use-Intensive Product Impact



Performing a DfE



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What aspects do you consider more important to include when identifying environmental impact?



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Design products and processes with materials that can be recycled continually with no loss in performance



Design products and process with natural materials that can be fully returned to the earth's natural cycle



Design products and processes that do not produce unnatural, toxic materials that are difficult to dispose of safely



Design products and processes with clean, renewable sources of energy rather than fossil fuels

Four
simple
DfE
rules

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Example: Herman Miller Setu chair



- Zero landfill
- Zero generation of hazardous waste
- Zero air emissions and process water use
- 100% use of green electrical energy
- 100% of sales from DfE products

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Example: Herman Miller Setu chair

- Environmentally friendly and non-toxic materials
 - 41% aluminum, 41% polypropylene, 18% steel, by weight
- Use of recycled materials
 - 44% by weight - 23% post-consumer, 21% post-industrial
- Less material content
 - 20 lbs lighter than most task chairs
- Easy to disassemble
 - 86% easily separable materials
- Recyclable
 - 92% by weight
- Production line uses 100% green power
- No air or water emissions released in production
- Returnable and recyclable packaging



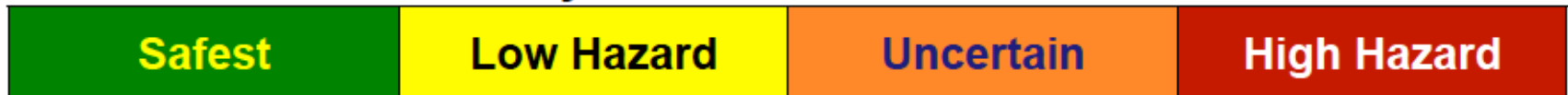
Source: Herman Miller, Inc.

 Herman Miller

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DfE assessment method for Herman Miller Setu chair

Material Chemistry (33.3%)



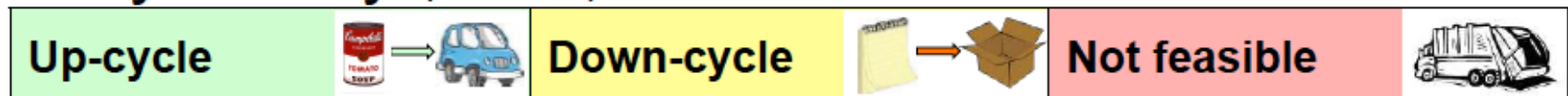
Recycled Content (8.4%)



Disassembly (33.3%)



Recyclability (25.0%)



Ref: Rossi, Charon, Wing, and Ewell, "Incorporating Cradle-to-Cradle Design into Herman Miller Products", *Journal of Industrial Ecology*, 2006.

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DfE assessment method for Herman Miller Setu chair

DfE Assessment Factor	Setu Score	Factor Weight	Weighted Score
Material Chemistry	50%	33.3%	16.7%
Recycled Content	44%	8.4%	3.7%
Disassembly	86%	33.3%	28.6%
Recyclability	92%	25%	23%
Overall Score		100%	72%

The overall DfE score for this product is high, @ 72%

How could it be improved further?

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What to include in the outline of a DfE?

- Decide which factors are the most important: materials? Use of sustainable routes? the ability of recycling after use?
- Select material: choose materials that show sustainable properties. If little information is available, make a judgment and select most promising option
- Select guidelines for DfE (see next slide)
- Note that the whole DfE process cannot be completed until an initial design is evaluated (out of scope for this module)

Design for Environment

Life Cycle Stage	Design for Environment Guidelines	
Materials	Sustainability of resources	- Specify renewable and abundant resources*. - Specify recyclable and/or recycled materials*. - Specify renewable forms of energy*.
	Healthy inputs and outputs	- Specify non-hazardous materials*. - Install protection against release of pollutants and hazardous substances. - Include labels and instructions for safe handling of toxic materials*.
Production	Minimal use of resources in production	- Employ as few manufacturing steps as possible*. - Specify materials that do not require surface treatments or coatings*. - Minimize the number of components*. - Specify lightweight materials and components*.
Distribution	Minimal use of resources in distribution	- Minimize packaging. - Use recyclable and/or reusable packaging materials. - Employ folding, nesting, or disassembly to distribute products in a compact state. - Apply structural techniques and materials to minimize the total volume of material.
Use	Efficiency of resources during use	- Implement default power down for subsystems that are not in use. - Use feedback mechanisms to indicate how much energy or water are being consumed. - Implement intuitive controls for resource-saving features.
	Appropriate durability	- Consider aesthetics and functionality to ensure the aesthetic life is equal to the technical life. - Facilitate repair and upgrading. - Ensure minimal maintenance. - Minimize failure modes.
Recovery	Disassembly, separation, and purification	- Ensure that joints and fasteners are easily accessible*. - Specify joints and fasteners so that they are separable by hand or with common tools*. - Ensure that incompatible materials are easily separated*.

DfE guidelines

(based on Telenko *et al*, 2008)

Summary



- DfE provides a more holistic and general view of the LCA of materials and products
- Outlining a process requires common knowledge of materials and processes
- The process can be built using little data however, details can be included if available
- The key to a good DfE outline is to include questions that allow to make a decision about design of material or process